

$$\bigcap_{i=1}^{\infty} A_i = (A_1 \cap A_2 \cap \dots \cap A_{\infty}) = \emptyset$$

c) $A_i = [-i, i]$, that is, the set of real numbers x with $-i \leq x \leq i$.

$$A_1 = [-1, 1] \quad A_2 = [-2, 2], \dots, A_i = [-i, i], \dots, A_{\infty} = [-\infty, \infty]$$

$$\bigcup_{i=1}^{\infty} A_i = A_1 \cup A_2 \cup \dots \cup A_i \cup \dots = [-\infty, \infty] = \mathbb{R}$$

$$\bigcap_{i=1}^{\infty} A_i = A_1 \cap A_2 \cap \dots \cap A_i \cap \dots = [-1, 1]$$

d) $A_i = [i, \infty)$, that is, the set of real numbers x with $x \geq i$.

$$\begin{aligned} \bigcup_{i=1}^{\infty} A_i &= A_1 \cup A_2 \cup \dots \cup A_i \cup \dots = [1, \infty) \cup [2, \infty) \cup \dots \cup [i, \infty) \cup \dots \\ &= [1, \infty) = A_1 \end{aligned}$$

$$\begin{aligned} \bigcap_{i=1}^{\infty} A_i &= A_1 \cap A_2 \cap \dots \cap A_i \cap \dots = [1, \infty) \cap [2, \infty) \cap \dots \cap [i, \infty) \cap \dots \\ &= \emptyset \end{aligned}$$

58) Suppose that the universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Express each of these sets with bit strings where the i th bit in the string is 1 if i is in the set and 0 otherwise.

a) $\{3, 4, 5\}$: 0011100000

b) $\{1, 3, 6, 10\}$: 1010010001

c) $\{2, 3, 4, 7, 8, 9\}$: 0111001110

59) Using the same universal set as in the last exercise, find the set specified by each of these bit strings.

a) 1111001111 $\rightarrow \{1, 2, 3, 4, 7, 8, 9, 10\}$

b) 0101111000 $\rightarrow \{2, 4, 5, 6, 7\}$

c) 1000000001 $\rightarrow \{1, 10\}$

61) What is the bit string corresponding to the difference of two sets?

Solution: $A - B = A \cap \bar{B}$ (Prob 21(a))

If the i th bit of A is 1 and B is 0, then the corresponding i th bit of $A \cap \bar{B}$ is 1. For the rest of the 3 combinations, the corresponding i th bit is 0. $\square$

62) What is the bit string corresponding to the symmetric difference of two sets?